

IoT Gateway BPC-iMX91-02 & BPC-iMX93-02 User Guide

Version: V1.0 (2026-06)

Compiled by: Polyhex Technology Company Limited (<http://www.polyhex.net/>)

The IoT Gateway BPC-iMX93-02 and BPC-iMX91-02 are rugged industrial computers based on the NXP i.MX93 and NXP i.MX91 processors, respectively. Each system consists of a SOM B (NXP iMX93) or SOM C (NXP iMX91) core board, a BMB-13-A1 carrier board, and a durable aluminum-alloy enclosure. Designed for reliable operation in demanding environments, the gateways feature rugged construction, dust resistance, vibration and shock resistance, wide-temperature support, and a compact, portable form factor.



Figure 1

REVISION HISTORY		
Rev.	Date	Description
1.0	2026.06.11	First edition

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Chapter 1 Security

1.1. Safety Precaution

This document informs how to make each cable connection. In most cases, you will simply need to connect a standard cable.

Table 1 Terms and conventions

Symbol	Meaning
<i>Warning!</i> 	Always disconnect the power cord from the chassis whenever there is no workload required on it. Do not connect the power cable while the power is on. A sudden rush of power can damage sensitive electronic components. Only experienced electricians should open the chassis.
<i>Caution!</i> 	Always ground yourself to remove any static electric charge before touching the product. Modern electronic devices are very sensitive to electric charges. Use a grounding wrist strap at all times. Place all electronic components on a static-dissipative surface or in a static-shielded bag.

1.2. Safety Instruction

To avoid malfunction or damage to this product please observe the following:

1. Disconnect the device from the DC power supply before cleaning. Use a cloth. Do not use liquid detergents or spray-on detergents.
2. Keep the device away from moisture.
3. During installation, set the device down on a reliable surface. Drops and bumps will lead to damage.
4. Before connecting the power supply, ensure that the voltage is in the required range, and the way of wiring is correct.
5. Carefully put the power cable in place to avoid stepping on it.

6. If the device is not used for a long time, power it off to avoid damage caused by sudden overvoltage.
7. Do not pour liquid into the venting holes of the enclosure, as this could cause fire or electric shock.
8. For safety reasons, the device can only be disassembled by professional personnel.
9. If any of the following situations occur, get the equipment checked by service personnel:
 - The power cord or plug is damaged.
 - Liquid has penetrated into the equipment.
 - The equipment has been exposed to moisture.
 - The equipment does not work well, or you cannot get it to work according to the user's manual.
 - The equipment has been dropped and damaged.
 - The equipment has obvious signs of breakage.
10. To prevent damage, do not place the device in any environment outside its specified operating temperature range. It needs to be kept in an environment at controlled temperature.
11. Due to the sensitive nature of the equipment, it must be stored in a restricted access location, only accessible by qualified engineer.

DISCLAIMER: Polyhex assumes no liability for the accuracy of any statement of this instructional document.

1.1. Technical Support

1. Visit DEBIX website <https://www.debix.io/> where you can find the latest information about the product.

■ Skip Links:

Debix Documentation: <https://debix.io/Document/manual.html>

Debix Blog: <https://debix.io/Software/blog.html>

Debix GitHub: <https://github.com/debix-tech>

2. Contact your distributor, sales representative or Polyhex's customer service center for technical support if you need additional assistance. Please have the following info ready before you call:

- Product name and memory size
- Description of your peripheral attachments
- Description of your software (operating system, version, application software, etc.)
- A complete description of the problem
- The exact wording of any error messages

■ TechSupport Platforms:

Discord Community (recommended): <https://discord.com/invite/adaHHaDkH2>

Email: teksupport@debix.io

Chapter 2 Introduction

The IoT Gateway BPC-iMX93-02 and BPC-iMX91-02 are RISC-based industrial gateway platforms featuring high performance, wide-temperature operation, and flexible hardware design. They serve as gateways between inverters and remote monitoring centers in power and energy applications, enabling reliable data acquisition, communication, and edge connectivity.

Main Features:

- **High-Performance Computing:** Powered by NXP i.MX9 series processors (i.MX91/i.MX93), delivering an optimal balance of performance, energy efficiency, and scalability for industrial IoT applications.
- **Comprehensive Connectivity:** Offers integrated Ethernet, Wi-Fi & Bluetooth, and optional cellular (4G) and long-range (LoRa) modules. UWB support enables precise location tracking.
- **Ruggedized Design:** Engineered to operate reliably in harsh environments, with excellent resistance to dust, shock, and vibration.
- **OS Support:** Supports both Yocto-L6.12.49 and Debian 13.

2.1. Overview

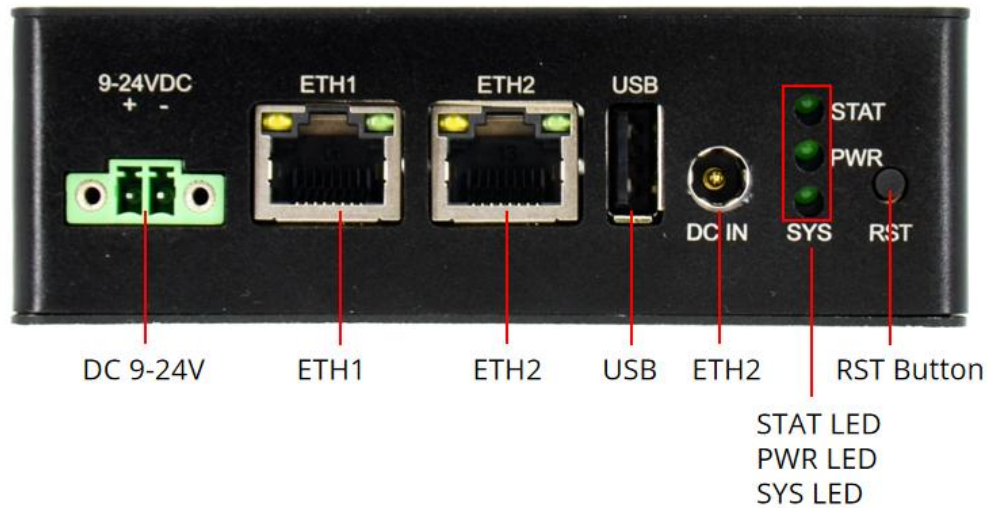


Figure 2 Front Interface

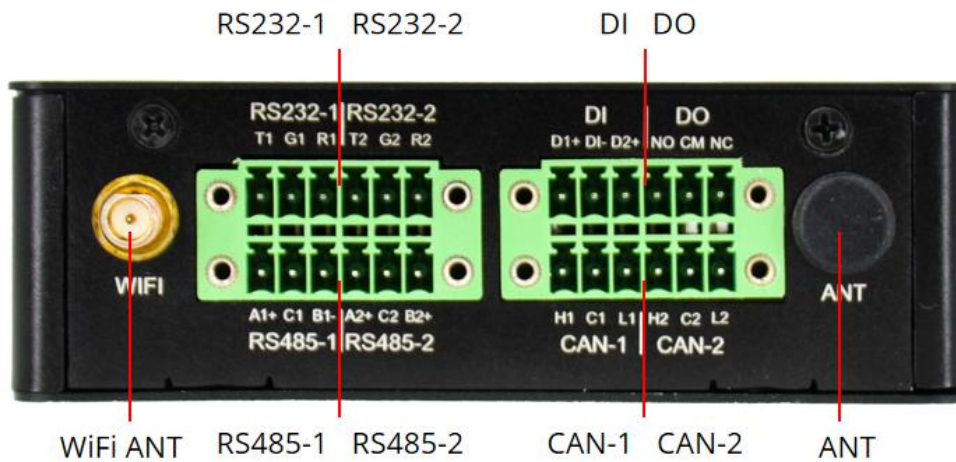


Figure 3 Rear Interface

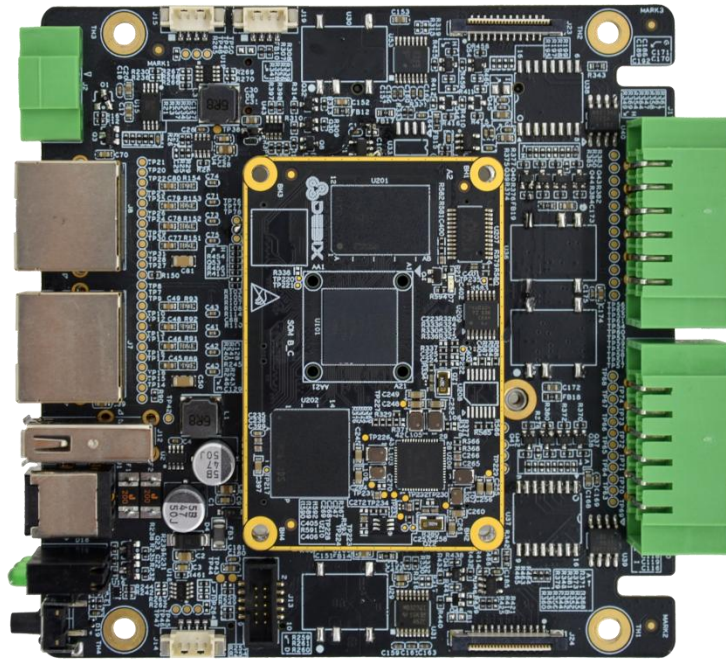


Figure 4 Top View of SOM B/SOM C core board + BMB-13-A1 carrier board

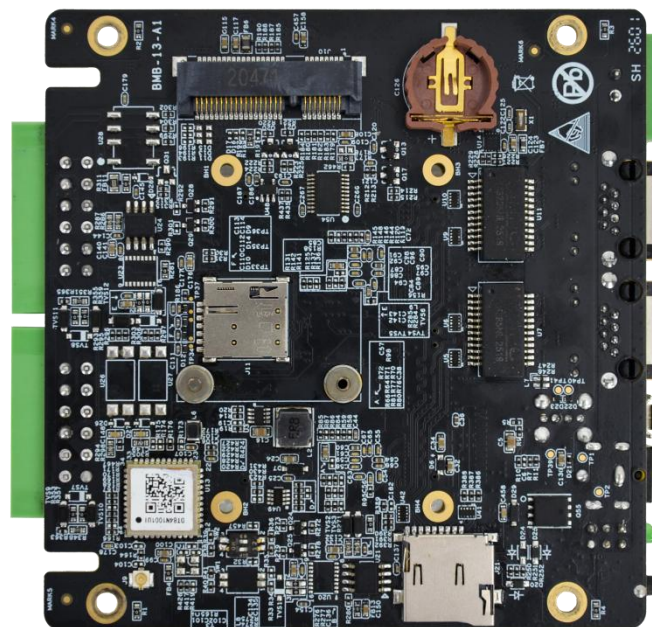


Figure 5 Bottom View of SOM B/SOM C core board + BMB-13-A1 carrier board

The IoT Gateway BPC-iMX93-02 and BPC-iMX91-02 are based on the SOM B (NXP i.MX93) or SOM C (NXP i.MX91) core board and the BMB-13-A1 carrier board. Both models support comprehensive connectivity options including Ethernet, 4G, LoRa, UWB, Wi-Fi, and Bluetooth. They feature a dustproof design and are resistant to shock and vibration, ensuring reliable operation in industrial environments. The detailed specifications are as follows.

Table 2 Specifications of BPC-iMX93-02 and BPC-iMX91-02

System		
Model	BPC-iMX93-02	BPC-iMX91-02
Main Board	SOM B core board + BMB-13-A1 carrier board	SOM C core board + BMB-13-A1 carrier board
CPU	NXP i.MX9352, 2 x Arm® Cortex®-A55 @1.7 GHz The max. PWR consumption is 1W	NXP i.MX9131, 1 x Arm® Cortex®-A55@1.4GHz. The maximum power consumption is 1.24W.
MCU	1 x Arm Cortex-M33 @250 MHz	None
GPU	2D Graphics Accelerator (PXP, Pixel Pipeline)	None
NPU	Arm® Ethos™ U-65 microNPU @0.5 TOPS	None
Security	Crypto, Tamper Detection, Secure Clock, Secure Boot, eFuse Key Storage, Random Number	
Watchdog	Hardware Watchdog	
Memory	1GB LPDDR4X (2GB optional)	
Storage	Onboard 8GB eMMC (16GB/32GB/64GB/128GB/256GB optional)	
OS	Yocto-L6.12.49, Debian 13	
Communication		
Gigabit Ethernet	1 x Gigabit Ethernet port (ETH1), supports TSN 1 x Gigabit Ethernet port (ETH2)	

4G	1 x 4G (optional)
LoRa	1 x LoRa (optional)
UWB	1 x UWB (optional)
Wi-Fi & Bluetooth	2.4GHz & 5GHz Wi-Fi 4, Bluetooth 5.2, 1 x external SMA Wi-Fi&BT antenna
External I/O	
USB	1 x USB 2.0 Host, Type-A interface (configurable as OTG)
LED	1) 1 x Power LED 2) 2 x GPIO LED (functions can be customized)
Reset	1 x Reset key
SMA RF	1 x Wi-Fi antenna (default), 1 x 4G/LoRa/UWB antenna (optional)
RS485	2 x RS485 (Non-isolated as standard, isolated optional)
RS232	2 x RS232 (Non-isolated as standard, isolated optional)
CAN	2 x CAN (Non-isolated as standard, isolated optional)
DIO	2 x DI, 2 x DO (Non-isolated as standard, isolated optional): ● Non-isolated DI (2-ch): Active-low, 3.3V, referenced to DIN_COM ● Non-isolated DO (2-ch): Active-low, 3.3V output ● Isolated DI (2-ch, optional): Active-high, external 5V supply required ● Isolated DO (2-ch, optional): Relay dry contact, max. 250VAC / 220VDC, 62.5VA (AC) / 60W (DC)
Internal I/O	
Slot	1 x NanoSIM card slot 1 x MicroSD card slot
Power Supply	
Power Input	Default DC 12V/2A power input, support DC 9V~24V wide voltage input 1) 1 x DC socket 2) 1 x 2Pin 3.5mm Phoenix terminals
Mechanical & Environmental	

Size (W x D x H)	107mm x 107mm x 35mm
Enclosure Material	Aluminum alloy
Net Weight	367g
Heat Dissipation	No fan, heat dissipation through the enclosure
Operating Temp.	-20°C~70°C (-40°C to 85°C optional)
Operating Humidity	20%~95%(non-condensing)
Storage Humidity	5%~95%(non-condensing)

2.2. External Interface

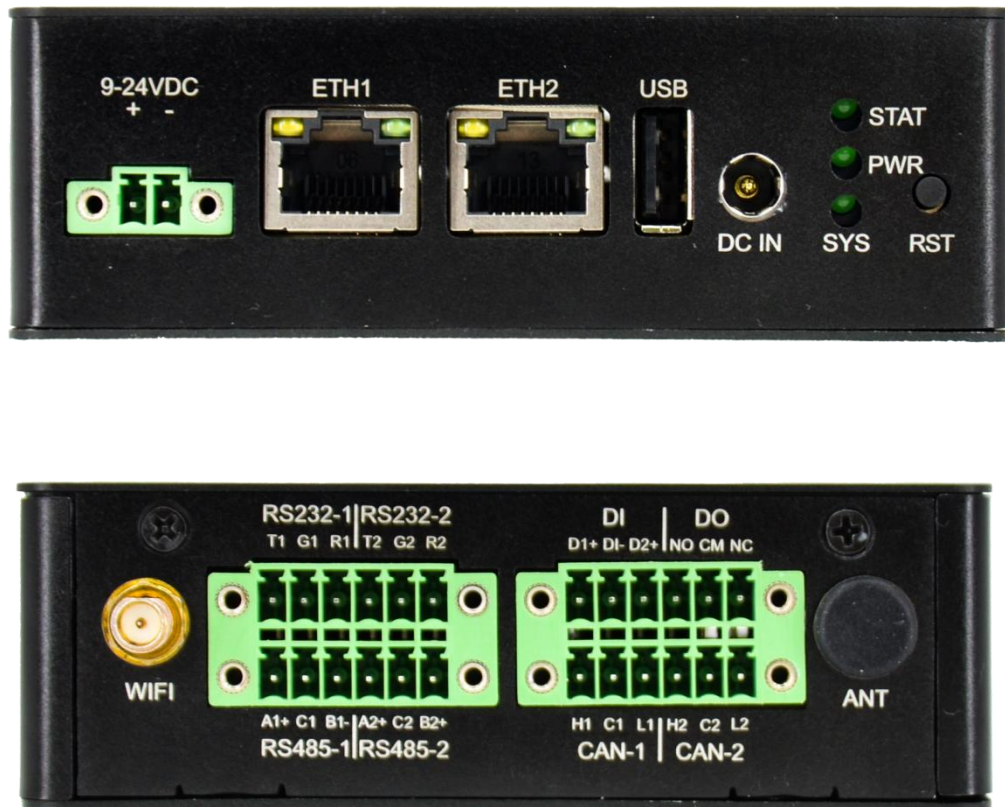


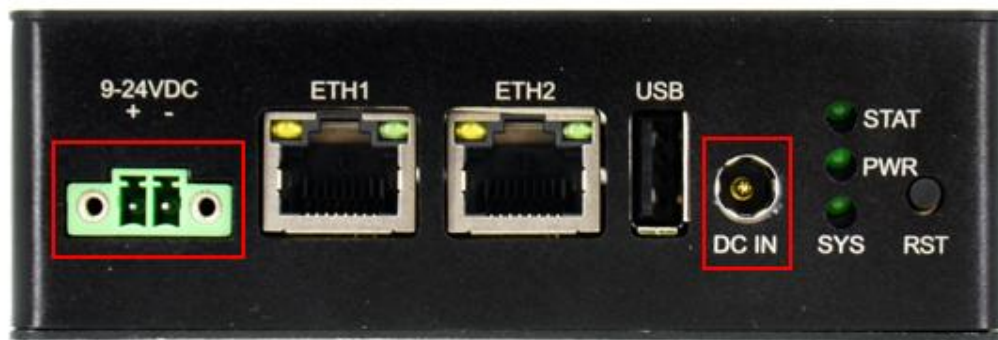
Figure 6 External I/O Interface

2.2.1. Power Interface

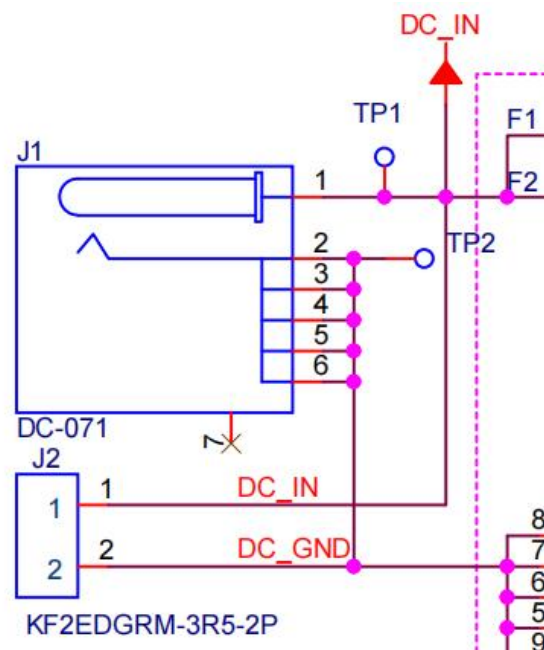
IOT gateway provides two power connectors with 9V~24V wide voltage input:

- 1 x 2Pin 3.5mm Phoenix terminals (J2)
- 1 x DC IN socket connector (J1)

As shown in the figure below.



The schematics is shown as below:



The Phoenix power connector (J2) is defined as follows:

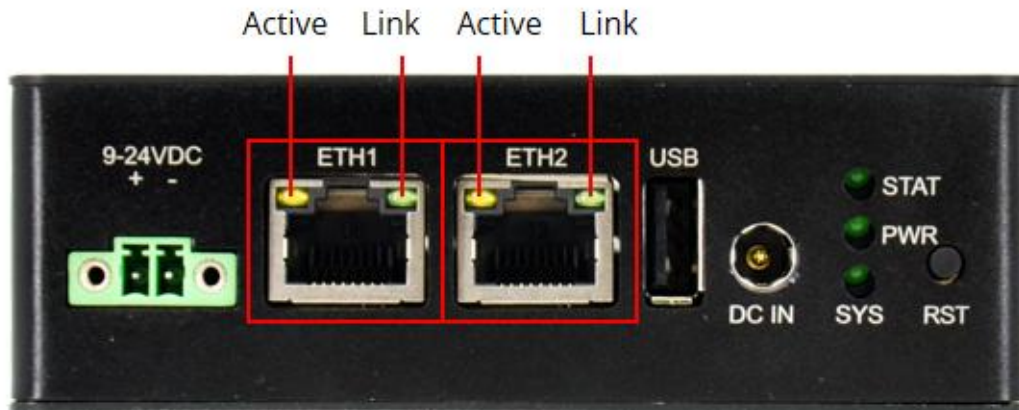
Table 3 Pin Definition of the Phoenix Power Connector (J2)

Pin	Definition	Description
1	+ (DC_IN)	DC power input positive pin, 9V~24V
2	- (DC_GND)	DC Power input negative pin

2.2.2. Ethernet Interface

IoT Gateway provides two independent MAC RJ45 Ethernet ports:

- Gigabit Ethernet1 (ETH-1 with TSN): the refdes. is J8
- Gigabit Ethernet2 (ETH-2): the refdes. is J7



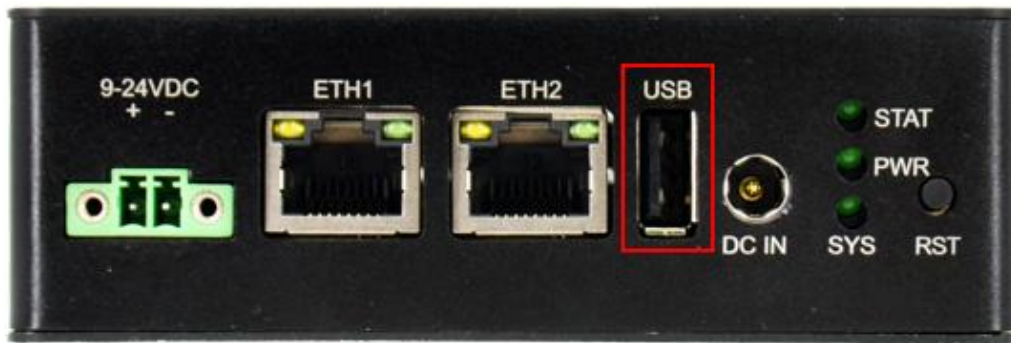
Connect the IOT gateway to the network through a network cable in the RJ45 connector. A set of status indicators above the interface displays the status signal. The green one indicates Link, which is network connection indicator, and the yellow one does Active, which is signal transmission indicator.

Table 4 Description of Ethernet Interface Status Indicator

LED	Color	Description
Link	Green	Light, the network cable is plugged in, network connection status is good
Active	Yellow	Blinking, network data is being transmitted

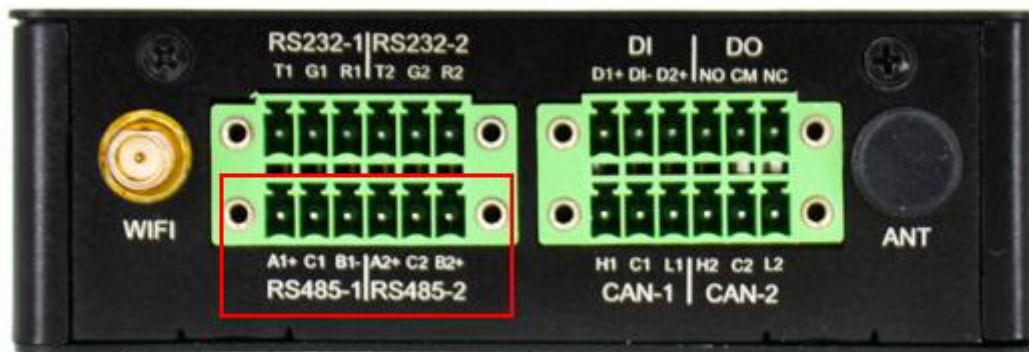
2.2.3. USB Interface

IoT Gateway provides a USB 2.0 host interface with Type-A connector, which can be configured as a USB OTG port.

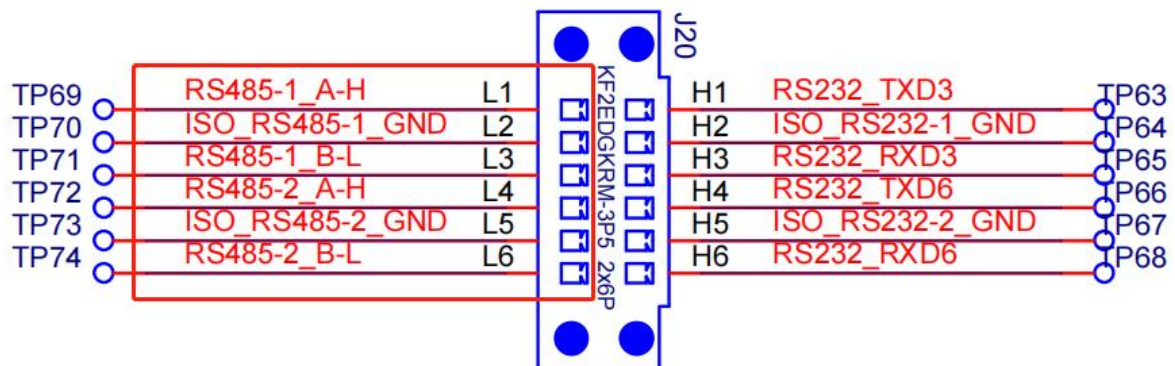


2.2.4. RS485 Interface

IoT Gateway provides two RS485 non-isolated interfaces by default via 3.5mm-pitch Phoenix terminals, with an option for physical isolation.



The pin sequence of two RS485 interfaces is shown as below:



The 2 x RS485 interfaces are defined as follows:

Table 5 Pin definition of RS485-1

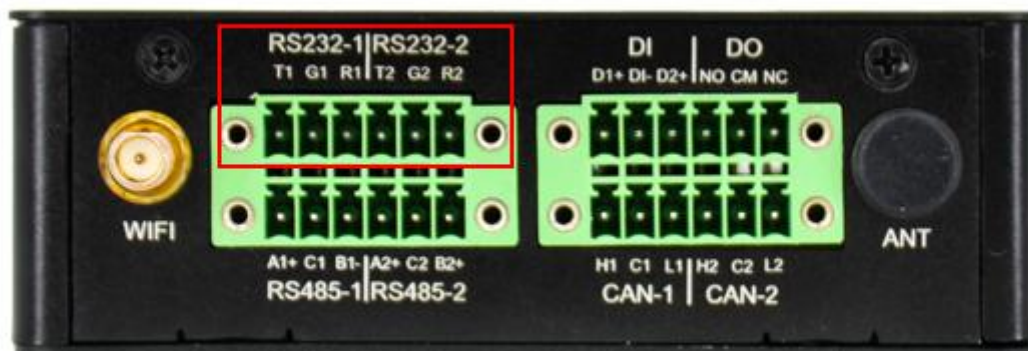
Pin	Definition	Description	Device node
L1	(A1+) RS485-1_A-H	Noninverting receiver input1 and Noninverting driver output1, 5V	/dev/ttyLP6
L2	(C1) ISO_RS485-1_GND	GND1	
L3	(B1-) RS485-1_B-L	Inverting receiver input1 and inverting driver output1, 5V	

Table 6 Pin definition of RS485-2

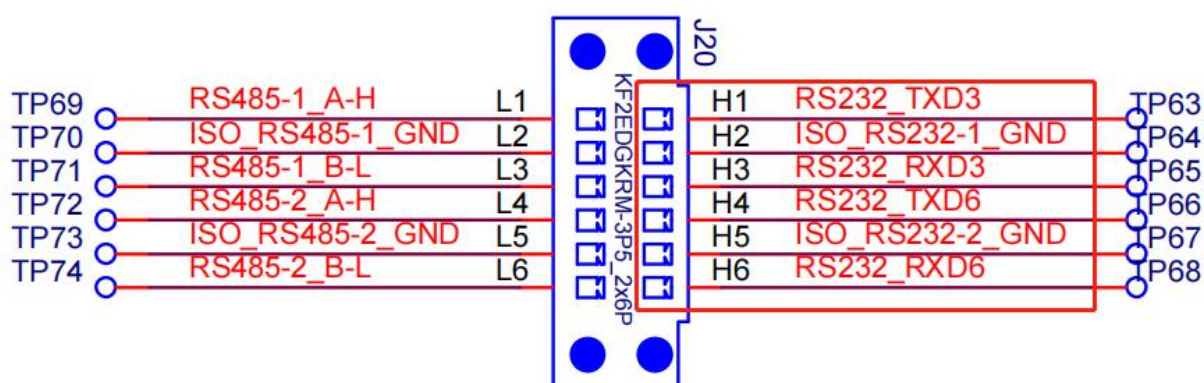
Pin	Definition	Description	Device node
L4	(A2+)RS485-2_A-H	Noninverting receiver input2 and Noninverting driver output2, 5V	/dev/ttyLP7
L5	(C2) ISO_RS485-2_GND	GND2	
L6	(B2+) RS485-2_B-L	Inverting receiver input2 and inverting driver output2, 5V	

2.2.5. RS232 Interface

IoT Gateway provides two RS232 non-isolated interfaces by default via 3.5mm-pitch Phoenix terminals, with an option for physical isolation.



The pin sequence of two RS232 interfaces is shown as below:



The 2 x RS232 interfaces are defined as follows:

Table 7 Pin definition of RS232-1

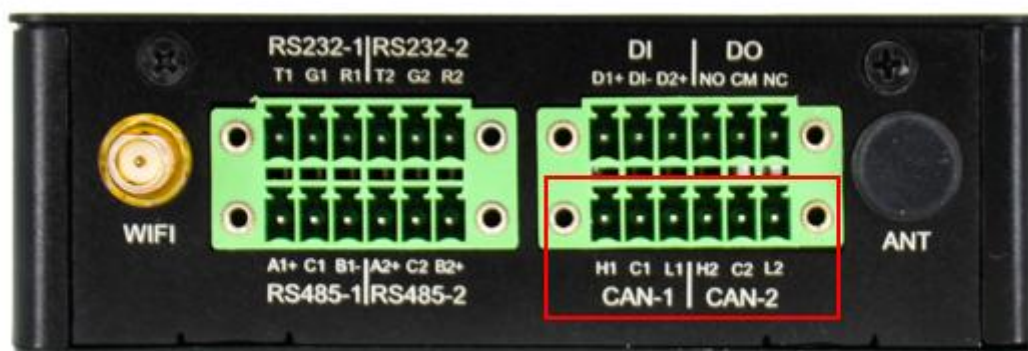
Pin	Definition	Description	Device node
H1	(T1) RS232_TXD3	Transfer data1, 5V	/dev/ttyLP2
H2	(G1) ISO_RS232-1_GND	GND1	
H3	(R1) RS232_RXD3	Receive data1, 5V	

Table 8 Pin definition of RS232-2

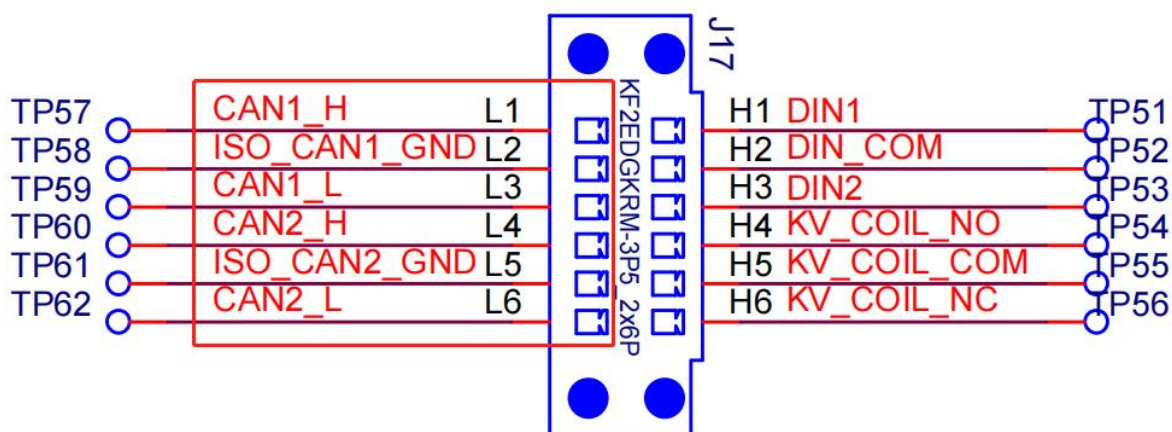
Pin	Pin	Description	Device node
H4	(T2) RS232_TXD6	Transfer data2, 5V	/dev/ttyLP5
H5	(G2) ISO_RS232-2_GND	GND2	
H6	(R2) RS232_RXD6	Receive data2, 5V	

2.2.6. CAN Interface

IoT Gateway provides two CAN non-isolated interfaces by default via 3.5mm-pitch Phoenix terminals, with an option of physical isolation.



The pin sequence of two CAN interfaces is shown as below:



The 2 x CAN interfaces are defined as follows:

Table 9 Pin definition of CAN-1

Pin	Definition	Description	Device node
L1	(H1) CAN1_H	High-level CAN bus line1, 5V	can0
L2	(C1) ISO_CAN1_GND	GND1	
L3	(L1) CAN1_L	Low-level CAN bus line1, 5V	

Table 10 Pin definition of CAN-2

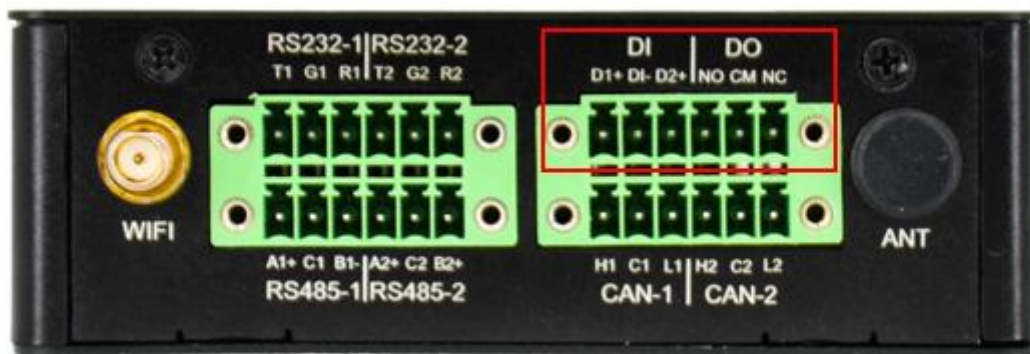
Pin	Definition	Description	Device node
L4	(H2) CAN2_H	High-level CAN bus line2, 5V	can1
L5	(C2) ISO_CAN2_GND	GND2	

L6	(L2) CAN2_L	Low-level CAN bus line2, 5V	
----	-------------	-----------------------------	--

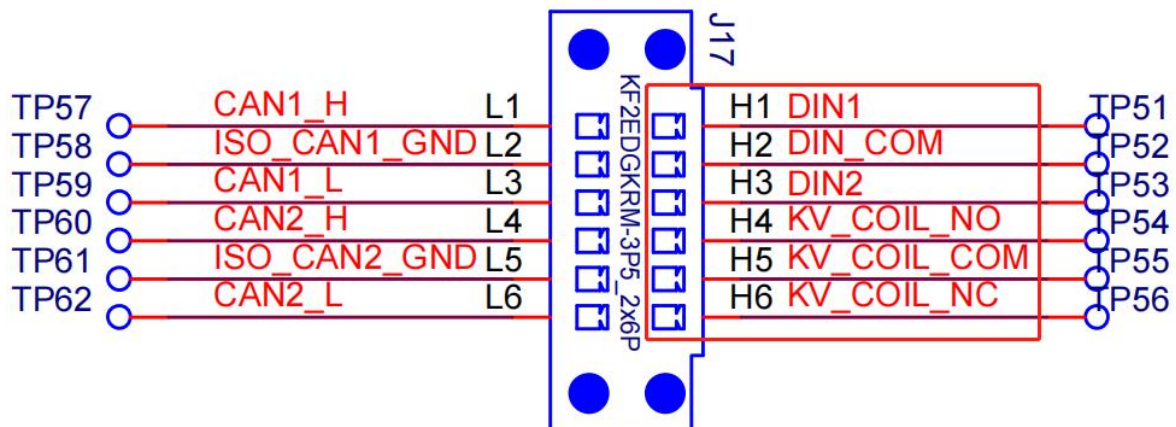
2.2.7. DI/DO Interface

IoT Gateway provides a group of DI and DO non-isolated interfaces by default via 3.5mm-pitch Phoenix terminals, with an option of physical isolation.

- Non-isolated DI (2-ch): Active-low, 3.3V, referenced to DIN_COM
- Non-isolated DO (2-ch): Active-low, 3.3V output
- Isolated DI (2-ch, optional): Active-high, external 5V supply required
- Isolated DO (2-ch, optional): Relay dry contact, max. 250VAC / 220VDC, 62.5VA (AC) / 60W (DC)



The pin sequence of the DI/DO interfaces is shown as below:



The DI/DO interfaces are defined as follows:

Table 11 Pin definition of DI

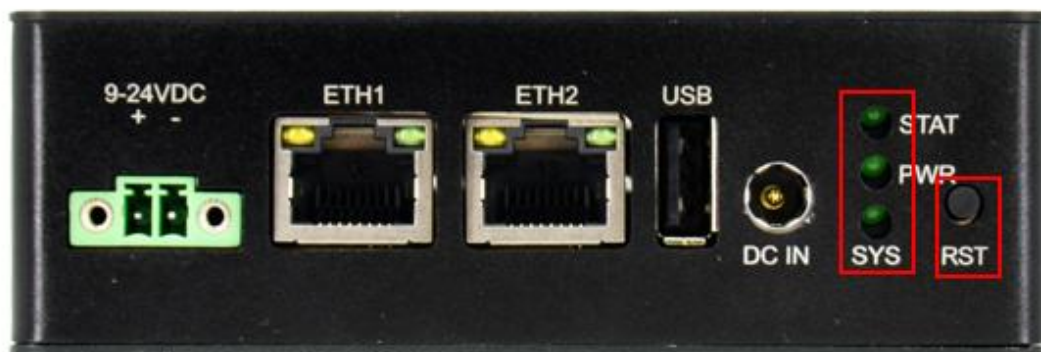
Pin	Definition	Description
H1	(D1+) DIN1	Digital input1 positive
H2	(DI-) DIN_COM	Digital input negative
H3	(D2+) DIN2	Digital input2 positive

Table 12 Pin definition of DO

Pin	Definition	Description
H4	(NO) KV_COIL_NO	Normal open
H5	(CM) KV_COIL_COM	Common
H6	(NC) KV_COIL_NC	Normal connected

2.2.8. LED and Key

IoT Gateway provides three LED status indicators and a Reset key, as shown in the figure below:



The specific status is described in the following table:

Table 13 Description of LED and Key

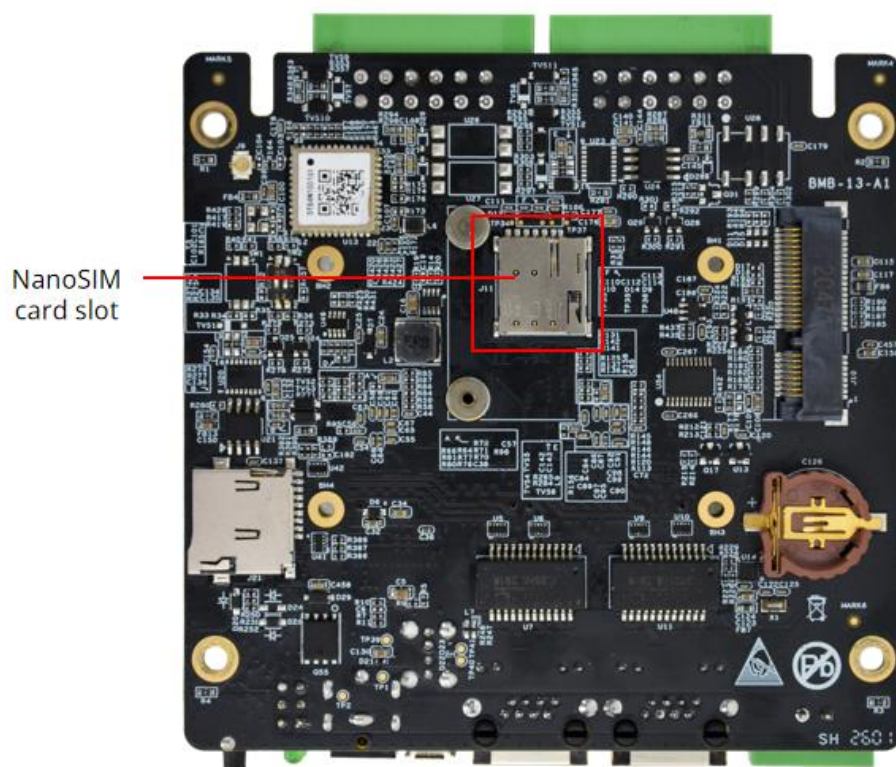
Function Name	Status	Description
---------------	--------	-------------

LED	STAT	Blinking	System works normally
	(controllable)	off	System works abnormally
	PWR	Blinking	Power is on
	(not controllable)	off	Power is off
	SYS	Lighting	Power is on
	(not controllable)	off	Power is off
Key	RST	Long press 8s	System reset

2.3. Internal Interface

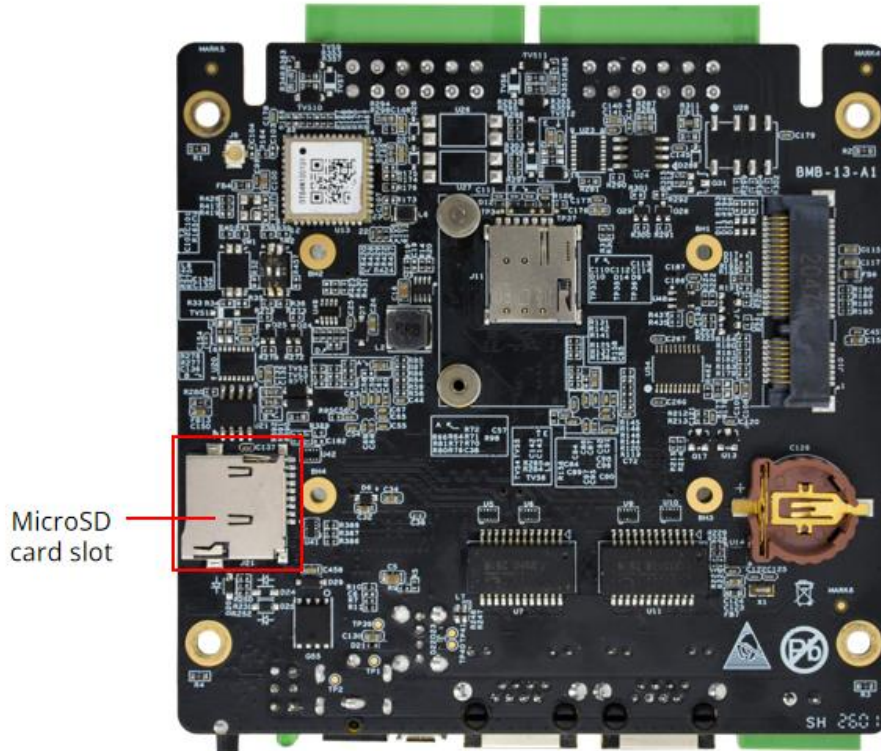
2.3.1. NanoSIM Card Slot

IoT Gateway provides a NanoSIM card slot on the PCB, as shown in the figure below:



2.3.2. MicroSD Card Slot

IoT Gateway provides a MicroSD card slot on the PCB as shown in the figure below.



2.4. Packing List

IoT gateway product list:

- 1 x BPC-iMX93-02 IOT Gateway / BPC-iMX91-02 IOT Gateway
- 1 x WiFi Antenna
- 1 x 2-pin Power Terminal Plug
- 2 x 12-pin Terminal Block Plugs

Chapter 3 Getting Started

3.1. Software Installation

IOT gateway is shipped with the operating system flashed into eMMC, and the system will boot from eMMC. If you need to reinstall the system, refer to the steps below.

3.1.1. USB Flash

- **Component Preparation**

- ✓ IOT Gateway (BPC-iMX93-02/BPC-iMX91-02)
- ✓ USB Type-A to Type-A OTG data cable
- ✓ DC 12V/2A Power Adapter
- ✓ PC (windows 10/11)

- **Download the Firmware Package**

- (1) BPC-iMX91-02 Firmware Package

- ✓ Download Link: <https://we.tl/t-fXwFjabZC5kyqeBf>
 - ✓ Package Name: iMX91_Firmware_Package.zip
 - ✓ SHA256:

- F81D508C1ED49AF968CD109772762E5AB88BD29FE9B0B904CEF831F0DE

1B692F

- (2) BPC-iMX93-02 Firmware Package

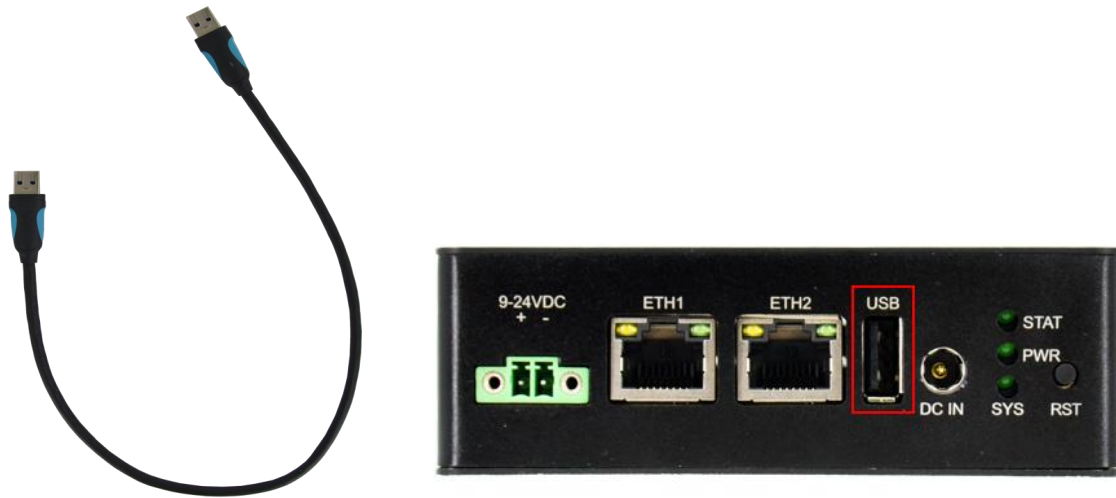
- ✓ Download Link: <https://we.tl/t-3i7Em7RZ6XmDUju7>
 - ✓ Package Name: iMX93_Firmware_Package.zip
 - ✓ SHA256:

- 8CA5A879CE15CFDFD44C7C9CC763F863EED48728205A425B4C04C2506E

1CDFC9

- **Flash to eMMC via USB**

1. Use a USB Type-A to Type-A OTG data cable to connect the USB OTG interface of the IOT gateway (BPC-iMX91-02/BPC-iMX93-02) and the USB interface of the PC.



2. Open the **Windows Command Prompt** as administrator on your PC.
3. Type the `cd` command to enter the root directory of the unzipped firmware package. For example:

```
cd D:\DEBIX_IOT_GATEWAY
```

4. Run the following command to download the file and start flashing the system to eMMC.

- BPC-iMX91-02

```
./uuu -b emmc_all imx-boot-imx91-bmb-13-a1-sd.bin-flash_singleboot
imx91-BMB-13-A1-v1.0.2-20260519.wic
```

- BPC-iMX93-02

```
./uuu -b emmc_all imx-boot-imx93-bmb-13-a1-sd.bin-flash_singleboot
imx93-BMB-13-A1-v1.0.1-20260511.wic
```

```
PS C:\Users\Administrator> cd C:\V1.0.2-20260519
PS C:\V1.0.2-20260519> .\uuu.exe -b emmc_all .\imx91-BMB-13-A1-v1.0.2-20260519.wic .\imx91-BMB-13-A1-v1.0.2-20260519.wic
uuu (Universal Update Utility) for nxp imx chips — libuuu_1.5.233-0-g79ce7d2
Wait for Known USB Device Appear...
```

5. Press and hold the RESET key on the IoT gateway while powering on the device. Release the key once a progress bar appears on the PC terminal, and then wait for the flashing process to complete. *(Note: Do not hold the RESET key for more than 8 seconds, as this will trigger a device reset.)*

```
PS C:\Users\Administrator> cd C:\V1.0.2-20260519
PS C:\V1.0.2-20260519> .\uuu.exe -b emmc_all .\imx-boot-imx91-bmb-13-a1-sd.bin-flash_singleboot .\imx91-BMB-13-A1-v1.0.2-20260519.wic
uuu (Universal Update Utility) for nxp imx chips — libuuu_1.5.233-0-g79ce7d2

Success 0   Failure 0

1:13-B347732 4/ 8 [ 4% ] FB: flash -raw2sparse all .\imx91-BMB-13-A1-v1.0.2-20260519.wic
```

- Wait for the system flashing to finish. When the terminal shows green “Done”, it means the flashing is finished.

```
PS C:\Users\Administrator> cd C:\V1.0.2-20260519
PS C:\V1.0.2-20260519> .\uuu.exe -b emmc_all .\imx-boot-imx91-bmb-13-a1-sd.bin-flash_singleboot .\imx91-BMB-13-A1-v1.0.2-20260519.wic
uuu (Universal Update Utility) for nxp imx chips — libuuu_1.5.233-0-g79ce7d2

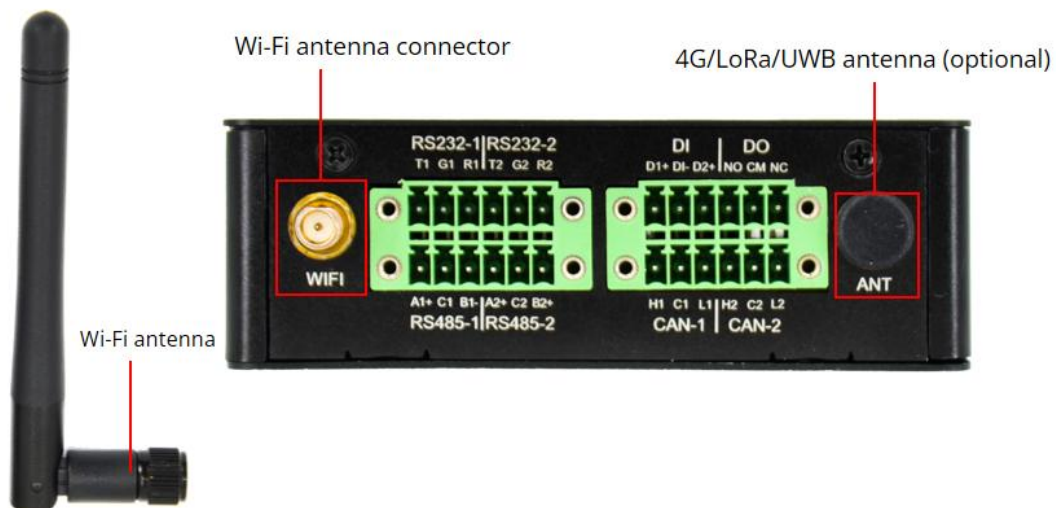
Success 1   Failure 0

1:13-B347732 8/ 8 [Done] ] FB: done
```

- After flashing, disconnect the power supply and OTG USB cable, and make sure the device is completely powered off. Then connect the power supply to start.

3.2. Antenna Installation

After receiving the product, install the Wi-Fi antenna to the Wi-Fi antenna connector as shown in the following figure.



NOTE

The factory default configuration of IoT Gateway BPC-IMX93-02 is Wi-Fi antenna.

If you need to configure other antennas, please contact our engineer for modification before leaving the factory.

3.3. Software Application Examples

3.3.1. Remote login SSH

Plug the cable into IoT Gateway network port, so that the device is connected to the LAN, enter the router background, query the IP address obtained by the device according to the MAC address, SSH to the device background through "putty" or other tools, access the account: root, the password is empty by default, as shown below:

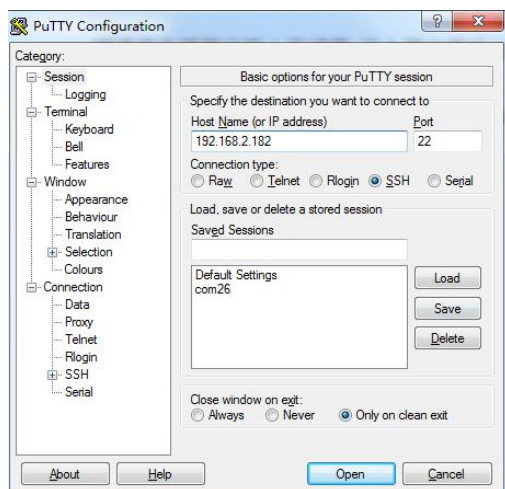


Figure 7

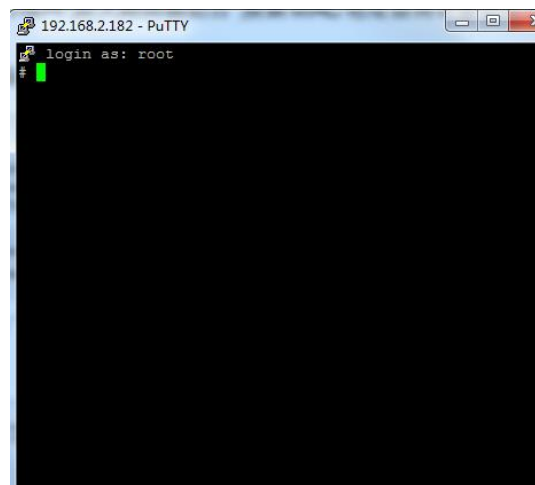


Figure 8

Change root password via `passwd root`, and enter the new password twice in a row.

3.3.2. Usage of Ethernet

1. Query ip command.

```
ip a
```

```
# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: can0: <NOARP,ECHO> mtu 16 qdisc noop state DOWN group default qlen 10
    link/can
3: can1: <NOARP,ECHO> mtu 16 qdisc noop state DOWN group default qlen 10
    link/can
4: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 10:07:23:6d:c6:12 brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.182/24 brd 192.168.2.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::1207:23ff:fe6d:c612/64 scope link
        valid_lft forever preferred_lft forever
5: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 10:07:23:6d:c6:13 brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.108/24 brd 192.168.2.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::1207:23ff:fe6d:c613/64 scope link
        valid_lft forever preferred_lft forever
6: sit0@NONE: <NOARP> mtu 1480 qdisc noop state DOWN group default qlen 1000
    link/sit 0.0.0.0 brd 0.0.0.0
7: wlan0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc pfifo_fast state DOWN group default qlen 1000
    link/ether ac:6a:a3:15:23:3f brd ff:ff:ff:ff:ff:ff
#
```

As shown above: eth0 network card corresponds to the network port of the device silkscreen "ETH1" (, left side);
eth1 network card corresponds to the network port of the device silkscreen "ETH2" (, right side).

2. Apply ping command.

```
ping -i eth0 192.168.2.254
```

```
# ping -i eth0 192.168.2.254
ping: option argument contains garbage: eth0
ping: this will become fatal error in the future
PING 192.168.2.254 (192.168.2.254) 56(84) bytes of data.
64 bytes from 192.168.2.254: icmp_seq=1 ttl=254 time=1.27 ms
64 bytes from 192.168.2.254: icmp_seq=2 ttl=254 time=0.733 ms
64 bytes from 192.168.2.254: icmp_seq=3 ttl=254 time=1.34 ms
64 bytes from 192.168.2.254: icmp_seq=4 ttl=254 time=1.19 ms
64 bytes from 192.168.2.254: icmp_seq=5 ttl=254 time=1.19 ms
64 bytes from 192.168.2.254: icmp_seq=6 ttl=254 time=1.17 ms
64 bytes from 192.168.2.254: icmp_seq=7 ttl=254 time=1.20 ms
64 bytes from 192.168.2.254: icmp_seq=8 ttl=254 time=1.17 ms
64 bytes from 192.168.2.254: icmp_seq=9 ttl=254 time=1.16 ms
64 bytes from 192.168.2.254: icmp_seq=10 ttl=254 time=1.19 ms
64 bytes from 192.168.2.254: icmp_seq=11 ttl=254 time=1.21 ms
64 bytes from 192.168.2.254: icmp_seq=12 ttl=254 time=1.23 ms
64 bytes from 192.168.2.254: icmp_seq=13 ttl=254 time=1.20 ms
64 bytes from 192.168.2.254: icmp_seq=14 ttl=254 time=1.23 ms
64 bytes from 192.168.2.254: icmp_seq=15 ttl=254 time=1.22 ms
64 bytes from 192.168.2.254: icmp_seq=16 ttl=254 time=1.23 ms
64 bytes from 192.168.2.254: icmp_seq=17 ttl=254 time=1.20 ms
64 bytes from 192.168.2.254: icmp_seq=18 ttl=254 time=1.23 ms
64 bytes from 192.168.2.254: icmp_seq=19 ttl=254 time=1.19 ms
```

3.3.3. Usage of WiFi

The following commands are used to configure and enable Wi-Fi connectivity on the device.

```
wpa_passphrase H3C_1CEC3C Polyhex.net_123 >> /etc/wpa_supplicant.conf
wpa_supplicant -i wlan0 -c /etc/wpa_supplicant.conf &
udhcpc -i wlan0
```

```
# udhcpc -i wlan0 -n
udhcpc: started, v1.34.1
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.168.10.8, server 192.168.10.254
udhcpc: lease of 192.168.10.8 obtained from 192.168.10.254, lease time 86400
deleting routers
adding dns 202.96.134.133
adding dns 114.114.114.114
#
```

3.3.4. Usage of Bluetooth

Start bluetooth and match bluetooth:

```
hciconfig hci0 up  
bluetoothctl  
  
power on  
  
agent on  
  
default-agent  
  
scan on  
  
pair yourDeviceMAC      #Match the Bluetooth MAC address
```

```
# bluetoothctl  
Agent registered  
[CHG] Controller AC:6A:A3:15:23:40 Pairable: yes  
[bluetooth]# power on  
Changing power on succeeded  
[bluetooth]# agent on  
Agent is already registered  
[bluetooth]# default-agent  
Default agent request successful  
[bluetooth]# scan on  
Discovery started  
[CHG] Controller AC:6A:A3:15:23:40 Discovering: yes  
[NEW] Device 6F:77:E4:55:30:6B 6F-77-E4-55-30-6B  
[NEW] Device 58:1F:3E:7C:17:CE 58-1F-3E-7C-17-CE  
[NEW] Device 61:8D:F0:19:75:3E 61-8D-F0-19-75-3E  
[NEW] Device 68:7A:15:E7:AD:CA 68-7A-15-E7-AD-CA  
[NEW] Device 78:21:08:79:5C:85 78-21-08-79-5C-85  
[NEW] Device 6F:66:07:AC:13:D7 6F-66-07-AC-13-D7  
[NEW] Device 68:E4:6A:8E:99:74 68-E4-6A-8E-99-74  
[NEW] Device 54:AF:B7:03:4D:69 54-AF-B7-03-4D-69  
[NEW] Device 74:5F:D2:47:FC:43 74-5F-D2-47-FC-43
```

```
[bluetooth]# pair 4C:02:20:3C:2A:6C
Attempting to pair with 4C:02:20:3C:2A:6C
[CHG] Device 4C:02:20:3C:2A:6C Connected: yes
Request confirmation
[agent] Confirm passkey 381184 (yes/no): yes
[CHG] Device 4C:02:20:3C:2A:6C Modalias: bluetooth:v038Fp1200d1436
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 00001105-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 0000110a-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 0000110c-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 00001112-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 00001115-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 00001116-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 0000111f-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 0000112f-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 00001132-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 00001200-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 00001800-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 00001801-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 0000fdaa-0000-1000-8000-00805f9b34fb
[CHG] Device 4C:02:20:3C:2A:6C UUIDs: 98b97136-36a2-11ea-8467-484d7e99a198
[CHG] Device 4C:02:20:3C:2A:6C ServicesResolved: yes
[CHG] Device 4C:02:20:3C:2A:6C Paired: yes
Pairing successful
```

3.3.5. Usage of USB

1. Access the U disk in FAT32 format, the system will automatically mount it to the /mnt path.

```
df -h
```

```
# df -h
Filesystem      Size      Used Available Use% Mounted on
/dev/root        4.8G      60.7M      4.4G      1% /
devtmpfs        163.9M          0      163.9M      0% /dev
tmpfs           244.4M          0      244.4M      0% /dev/shm
tmpfs           244.4M      68.0K      244.4M      0% /tmp
tmpfs           244.4M      40.0K      244.4M      0% /run
/dev/mmcblk1p3   1.7G       60.0K       1.6G      0% /recovery
/dev/sda1        28.7G     544.0K      28.7G      0% /mnt
#
```

- If the U disk is not mounted, you can mount the U disk with the following command:
 - Query the U disk letter:

```
fdisk -l
```

```
# fdisk -l
Disk /dev/mmcblk1: 7456 MB, 7818182656 bytes, 15269888 sectors
238592 cylinders, 4 heads, 16 sectors/track
Units: sectors of 1 * 512 = 512 bytes

Device      Boot  StartCHS      EndCHS          StartLBA        EndLBA        Sectors   Size Id Type
/dev/mmcblk1p1  320,0,1      959,3,16         20480         1044479        1024000   500M  c Win95 FAT32 (LBA)
/dev/mmcblk1p2  768,0,1      639,3,16        1228800       11509759       10280960  5020M 83 Linux
/dev/mmcblk1p3  640,0,1      1023,3,16        11509760      15269887       3760128  1836M 83 Linux
Disk /dev/sda: 29 GB 30784094208 bytes, 60125184 sectors
3742 cylinders, 255 heads, 63 sectors/track
Units: sectors of 1 * 512 = 512 bytes

Device      Boot  StartCHS      EndCHS          StartLBA        EndLBA        Sectors   Size Id Type
/dev/sda1    0,0,33       1023,254,63      32             60125183       60125152  28.6G  c Win95 FAT32 (LBA)
#
```

- Mounting the U disk:

```
mount /dev/sda1 /mnt
```

```
# mount /dev/sda1 /mnt
# df -h
Filesystem      Size      Used Available Use% Mounted on
/dev/root        4.8G      60.8M      4.4G      1% /
devtmpfs        163.9M          0      163.9M      0% /dev
tmpfs           244.4M          0      244.4M      0% /dev/shm
tmpfs           244.4M      52.0K      244.4M      0% /tmp
tmpfs           244.4M      36.0K      244.4M      0% /run
/dev/mmcblk1p3   1.7G       60.0K       1.6G      0% /recovery
/dev/sda1        28.7G     544.0K      28.7G      0% /mnt
#
```

2. Enter the U disk directory:

```
cd /mnt
```

```
# cd /mnt/
# ls
BMB07-factorytest  BMB07-factorytest-V1.0.4.rar  System Volume Information
# cp BMB07-factorytest-V1.0.4.rar /root/
```

3.3.6. Verification of RS232

Connect T1 of RS232-1 to the receiving end R2 of RS232-2, R1 to the sending end T2 of RS232-2, and G1 to the ground terminal G2 of RS232-2. The wiring is shown in the following figure.

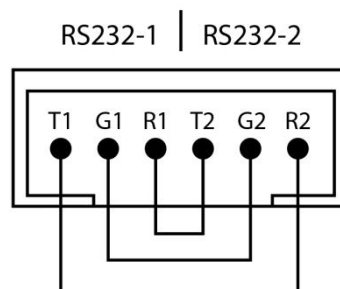


Figure 9

- Set RS232-1 to write data and RS232-2 to read data

```
uart_read /dev/ttyLP5 115200 &          #(RS232-2 backend read data)
uart_write /dev/ttyLP2 115200 123        #(RS232-1 write data)
uart_write /dev/ttyLP2 115200 123456     #(RS232-1 write data)
killall uart_read
```

- Switch to RS232-2 to write data, RS232-1 to read data

```
uart_read /dev/ttyLP2 115200 &          #(RS232-1 backend read data)
uart_write /dev/ttyLP5 115200 456        #(RS232-2 write data)
uart_write /dev/ttyLP5 115200 456789     #(RS232-2 write data)
killall uart_read
```

3.3.7. Verification of RS485

Connect A1 to A2, and connect B1 to B2, and connect C1 to C2. The wiring is shown in the following figure.

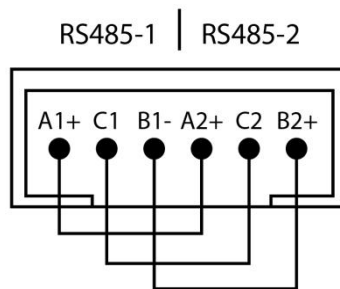


Figure 10

The RS485 control signals (RS485_nRE1, RS485_nDE1, RS485_nRE2, and RS485_nDE2) are set to a high level by default.

- The signal levels can be controlled using the following commands:

```
echo 1 > /sys/class/leds/RS485_nRE1/brightness
```

```
echo 0 > /sys/class/leds/RS485_nRE1/brightness
```

```
echo 1 > /sys/class/leds/RS485_nRE2/brightness
```

```
echo 0 > /sys/class/leds/RS485_nRE2/brightness
```

```
echo 1 > /sys/class/leds/RS485_nDE1/brightness
```

```
echo 0 > /sys/class/leds/RS485_nDE1/brightness
```

```
echo 1 > /sys/class/leds/RS485_nDE2/brightness
```

```
echo 0 > /sys/class/leds/RS485_nDE2/brightness
```

echo 1 sets the corresponding signal to a high level.

echo 0 sets the corresponding signal to a low level.

- Set RS485-2 to write data and RS485-1 to read data

```
uart_read /dev/ttyLP6 115200 &          #(RS485-1 backend read data)
```

```
uart_write /dev/ttyLP7 115200 123        #(RS485-2 write data)
```

```
uart_write /dev/ttyLP7 115200 123456     #(RS485-2 write data)
```

```
killall uart_read
```

- Switch to RS485-1 to write data, RS485-2 to read data

```
uart_read /dev/ttyLP7 115200 &          #(RS485-2 backend read data)
```

```
uart_write /dev/ttyLP6 115200 456        #(RS485-1 write data)
```

```
uart_write /dev/ttyLP6 115200 456789     #(RS485-1 write data)
```

```
killall uart_read
```

3.3.8. Verification of CAN

Connect H1 to H2, and connect L1 to L2 (that is, H to H, and L to L). The wiring is shown in the following figure.

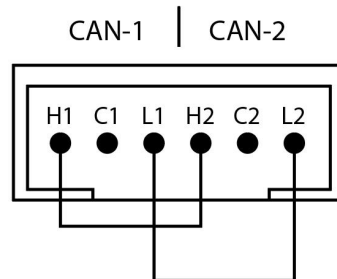


Figure 11

- CAN1 sends data, CAN0 receives data

Open a Terminal, configure CAN0 to receive, and CAN1 to send.

```
ifconfig can0 down
ifconfig can1 down
ip link set can0 type can bitrate 500000
ip link set can1 type can bitrate 500000
ifconfig can0 up
ifconfig can1 up
candump can0 &                                #(can0 backend receive data)
cansend can1 123#1122334455667788             #(can1 send data)
```

- CAN1 receives data, CAN0 sends data

In the Terminal, switch can1 to receive and can0 to send.

```
candump can1 &                                #(can1 backend receive data)
cansend can0 123#1122334455667788             #(can0 send data)
```

3.3.9. Verification of DI

IoT Gateway provides two DI non-isolated interfaces by default via 3.5mm-pitch Phoenix terminals, with an option of physical isolation.

- Non-isolated DI (2-ch): Active-low, 3.3V, referenced to DIN_COM
- Isolated DI (2-ch, optional): Active-high, external 5V supply required
- DI Testing Method

The DI functionality can be verified using the `debix_Din_test` utility.

- Behavior Description
 - Isolated DI version:
Inputs are active-high. Signal transitions will be detected as UP / DOWN events.
 - Non-isolated DI version:
Inputs are active-low. Signal transitions will also be detected as UP / DOWN events.

```
root@Debix:~# debix_Din_test
Din1 DOWN
Din1 UP
Din2 DOWN
Din2 UP
```

3.3.10. Verification of DO

IoT Gateway provides two DO non-isolated interfaces by default via 3.5mm-pitch Phoenix terminals, with an option of physical isolation.

- Non-isolated DO (2-ch): Active-low, 3.3V output
- Isolated DO (2-ch, optional): Relay dry contact, max. 250VAC / 220VDC, 62.5VA (AC) / 60W (DC)
- DO Testing Method
 - Non-isolated DO:

Control the output state using the following commands:

```
echo 0 > /sys/class/leds/Dout1/brightness
echo 1 > /sys/class/leds/Dout1/brightness
echo 0 > /sys/class/leds/Dout2/brightness
echo 1 > /sys/class/leds/Dout2/brightness
```

`echo 0`: Drive the output low

echo 1: Drive the output high

- Isolated DO:

Control the relay state using the following commands:

```
echo 0 > /sys/class/leds/KV_COIL_EN/brightness
echo 1 > /sys/class/leds/KV_COIL_EN/brightness
```

echo 0: De-energize the relay coil

echo 1: Energize the relay coil

The relay output provides both Normally Open (NO) and Normally Closed (NC) contacts. The contact state changes according to the relay coil status.

3.3.11. Usage of 4G Module

- 4G Connectivity Test

Run the following script to establish a cellular data connection:

```
debix_4g_connect.sh
```

After the dial-up connection is established successfully, check the network interface using:

```
ip a
```

A ppp0 interface with an assigned IP address should be displayed.

- GPS Test

- Prerequisites

Ensure that GPS support and gpsd have been installed. Copy the GPSD configuration file:

```
cp /etc/default/gpsd.default /etc/default/gpsd
```

- Enable GPS

Enable the GNSS function of the Quectel module:

```
echo -e "AT+QGPS=1\r" > /dev/ttyUSB2
```

- Start the GPSD service:

```
systemctl start gpsd
```

- Verify GPS Operation

Run the GPS test utility:

```
debix_gps_test
```

Alternatively, use: `gpsmon` or `cgps -s`

- Expected Result

Once valid satellite signals are received, GPS information such as time, latitude, and longitude will be displayed. Example:

```
Time: 2026-05-07T12:00:46.000Z   Lat: 22 35.455214' N   Lon: 113 55.428324' E |
----- Cooked TPV -----
GPGLV GPGGA GPRMC GPVTG GPRMC GPGSA
----- Sentences -----
```

3.3.12. Verification of RTC

- Confirm that the pcf85363 driver module is loaded successfully.

```
dmesg | grep pcf85363
```

- Read the RTC time.

```
hwclock --systohc
```

```
hwclock --show
```

```
root@Debix:~# hwclock -w
root@Debix:~# hwclock -r
2025-05-29 16:33:22.884353+00:00
root@Debix:~#
```